

ATLIQ MEDIA · CODEBASICS RPC #21

Tamil Nadu Assembly Election 2026

AtliQ Media — Neutral Election Briefing

VISUAL & REFERENCE MASTER

ECI public data only · 234 constituencies · 2021 vs 2026

Generated: May 28, 2026 · Descriptive analysis · No predictions, no causal claims

8 · VISUAL & REFERENCE MASTER

Visual & Reference Master

Visual Design & Reference Master Guide

Current submission stack (May 2026): React (`web/`) + FastAPI (`api/`) + ECharts. Run `uvicorn api.main:app --port 8000` (after `cd web && npm run build`) or `npm run dev` on `:5173`. Sections that mention **Streamlit** or `dashboard/app.py` are **design references only** — not the shipped RPC dashboard.

Deep insights page now live (`/deep`): Effective Number of Parties (Laakso–Taagepera), Pedersen volatility, Gallagher LSq disproportionality, vote-share swing heatmap, anti-incumbency, representation gap, race competitiveness, district churn. All descriptive, ECI-only. Definitions in `docs/METRIC_DEFINITIONS.md`.

Purpose: Single reference for building a **world-class, neutral, creative** Tamil Nadu 2026 election story — not limited to Codebasics rules. Use this alongside `HACKATHON_EXECUTION_PLAN.md` for submission logistics.

How to use: Pick **one headline story** → choose **3 chart types** from Section 8 → apply **accessibility + ethics** (Sections 9–10) → implement with **tech stack** (Section 11) → validate against **data sources** (Section 12).

1. Reference ecosystem (who to learn from)

```

flowchart TB
  subgraph challenge [Your deliverables]
    V[Video pitch]
    D[Stakeholder deck]
    G[GitHub notebook]
    DB[Interactive dashboard]
  end

  subgraph global [Global newsroom patterns]
    BBC[BBC India 2024 maps]
    NPR[NPR tile maps + live results]
    FT[FiveThirtyEight / 538 flows]
    AP[AP VoteCast Sankey]
    Vox[Vox cartogram critique]
  end

  subgraph india [India public examples]
    Elexa[News18 Elexa]
    Quint[Quint Data Studio]
    Hindu[The Hindu infographics]
    IIT[IIT Bombay Election Explorer]
    DM[Datameet maps + data]
  end

  subgraph creative [Creative / scrolllytelling]
    Pudding[The Pudding visual essays]
    IIB[Information is Beautiful Awards]
  end

  subgraph standards [Standards and ethics]
    RSS[RSS DataVis Guide]
    EU[EU Data Viz Accessibility]
    APeth[AP factual reporting]
    RPC[Codebasics RPC neutrality]
  end

  global --> DB
  india --> DB
  creative --> V
  creative --> D
  standards --> challenge
  RPC --> challenge

```

2. Codebasics RPC (mandatory constraints)

Item	Requirement
Deadline	28 May 2026, 11:59 PM IST
Scoring	~70% storytelling (video + deck), ~30% data hygiene
Data	ECI public + starter CSVs only
Tone	No causal claims, predictions, party praise/criticism
Visuals	No leader photos, party symbols
Deliverables	Video 5–7 min → Deck 8–10 slides → GitHub → Dashboard optional
Research	Pick 3 of 6 questions; weave one story

Full detail: [HACKATHON_EXECUTION_PLAN.md](#), [How_to_Submit.docx](#), [metadata.txt](#).

3. Global newsroom dashboards — patterns to adopt

3.1 BBC News — India 2024 election results

- **URL:** [BBC India 2024 results \(interactive\)](#)
- **Patterns:** Constituency-level drill-down; **year-over-year compare** (2024 vs 2019); battleground highlights; search; map scaled for **readability over geographic truth**
- **Steal for TN:** Compare **2026 vs 2021** at AC level; default to **regional summary** then drill to constituency

3.2 NPR — Election results & tile maps

- **Live results:** [NPR 2024 election apps](#)
- **Tile maps:** [NPR hex/tile map methodology](#)
- **Patterns:** Don't over-color incomplete counts; **uniform tiles** so rural ACs aren't visually erased; sidebar filters
- **Steal for TN:** **234-tile mosaic** grouped by 6 editorial regions (not geographic choropleth if no clean shapefile)

3.3 Associated Press — AP VoteCast

- **URL:** [AP VoteCast 2024 \(IIB showcase\)](#)
- **Patterns:** **Sankey** for flows between groups; scan path for casual users; deep tables for power users
- **Steal for TN:** **2021 winner party → 2026 winner party** Sankey (seat flows, not voter demographics)

3.4 FiveThirtyEight / 538

- **Forecast UX:** [538 2020 forecast \(IIB\)](#)
- **Demographics:** [Swing-O-Matic style tools](#)
- **Patterns:** Uncertainty as first-class citizen; plain-language “robotext”; accessibility audits; **avoid** implying prediction for RPC
- **Steal for TN:** **Margin distributions** and “how fragmented is the mandate” — descriptive only, no forecast

3.5 Vox & National Geographic — map literacy

- **Vox:** [Why election maps mislead](#)
- **NatGeo:** [Cartogram alternatives](#)
- **Lesson:** Choropleths overweight land area; cartograms cost geography literacy; **tile grids** are the compromise
- **Steal for TN:** If you use a map, add **bar chart of same data** side-by-side (dual encoding)

3.6 ICA research — map type affects accuracy

- **Source:** Copernicus ICA work on election map types (choropleth vs cartogram vs block maps)
 - **Lesson:** Saturated reds/blues increase perceived partisanship; **desaturate**; accent only key views
 - **Steal for TN:** Use **muted Okabe–Ito** palette; never saffron/green dyads
-

4. India — public dashboards & data hubs

Source	URL	What to study
ECI results portal	https://results.eci.gov.in/	Official 2026 TN live results; turnout if missing in CSV
ECI statistical reports	https://www.eci.gov.in/statistical-reports	PDF validation, historical context
TN CEO	https://elections.tn.gov.in/	State Form-20, master lists
data.gov.in (ECI)	https://data.gov.in	Downloadable CSV/Excel
News18 Elexa	https://www.news18.com/elections/analytics-center/	Filters, grid/list/map, search
Quint Data Studio	https://www.thequint.com/quintlab/	Live seat maps, constituency explorer
The Hindu infographics	e.g. Bihar 2025 results pages	Clean small-multiples, filterable charts
IIT Bombay Election Explorer	https://info-design-lab.github.io/General-Election-Explorer/	Cartogram + map; ECI + Datameet
Parliament 2009–2024 explorer	https://data-analytics.github.io/Election_Data_2024/	Multi-cycle compare UX
IndiaVotes	https://indiavotes.com	Cross-check constituency numbers
Datameet election data	https://github.com/datameet/india-election-data	Open historical datasets
Datameet AC shapefiles	https://github.com/datameet/maps/tree/master/assembly-constituencies	Choropleth (verify delimitation caveats)
Trivedi Centre	Ashoka / TCPD (2021 TN source in metadata)	Academic cleaning standards

TN-specific note: Constituency boundaries on Datameet may be **pre-delimitation** for some states — document in limitations if used.

5. Creative & award-winning visual journalism

5.1 The Pudding — visual essays

- **Method:** [How Pudding structures visual essays](#)
- **Scrollytelling:** [Scrollytelling examples](#)
- **Patterns:** Pinned chart + scrolling narrative; sparse words; data carries conclusion
- **Steal for RPC: Video** = spoken scrollytelling; deck slide 1 = “pinned” headline chart

5.2 Information is Beautiful Awards

- Browse: <https://www.informationisbeautifulawards.com/> (Current Affairs & Politics)
- **Steal:** One **hero visual** (Sankey or mosaic) that works without reading body text

5.3 Other high-impact formats (industry)

Format	Best for	Example use in TN RPC
Sankey / alluvial	Party seat flows 2021→2026	Q2 Flips
Tile / waffle grid	234 ACs equal visual weight	Q1 Geographic + explorer
Sunburst / treemap	Region → party hierarchy	Regional seat composition
Slope / bump chart	Rank or metric change by entity	Margin 2021 vs 2026 by AC sample
Beeswarm / strip	Distribution of margins	Q6 fragmentation
Marimekko	Vote share composition by region	Q3 TVK footprint (descriptive)
Chord diagram	Flow matrix (use sparingly — hard to read)	Alternative to Sankey
Parallel coordinates	Multi-metric AC profiles	Power-user dashboard tab
Small multiples	6 regions same chart type	Q1
Dumbbell chart	Two-year comparison	Turnout delta (if scraped)

6. Accessibility & inclusive design (required for “best” work)

Standard	URL	Apply
EU Data Visualization Guide	https://data.europa.eu/apps/data-visualisation-guide/	Contrast $\geq 4.5:1$ text; don't use color alone
RSS Best Practices	https://royal-statistical-society.github.io/datavisguide/	Direct labels, alt text, chart choice
Urban Institute — Do No Harm	https://www.urban.org/research/publication/do-no-harm-guide	Center accessibility from day one
CT Open Data guidelines	https://ctopendata.github.io/data-visualization-guidelines/	Audience, goals, tables alongside charts
Okabe-Ito palette	Standard colorblind-safe set	Default party-agnostic colors
WCAG 2.x	W3C	Keyboard nav for web dashboard; semantic HTML

Checklist before ship:

- ☐ No red–green only encoding
- ☐ Labels on lines/bars (not legend-only)
- ☐ Tooltips + **data table** export for each main chart
- ☐ Alt text / short description for deck stills
- ☐ Font size ≥ 12 pt equivalent on TV-safe deck charts

7. Neutral reporting & ethics (aligns with RPC disqualification rules)

Source	Principle
AP Elections	Report facts; explain methodology; no “likely winner” without math path
ACE Project (media ethics)	Separate fact from opinion; balance; label commentary
RPC brief	No causal “why party won”; no predictions; ECI-only

Safe chart title templates:

- ☒ “108 constituencies had TVK as leading candidate in published results”
- ☒ “TVK destroyed DMK in the south”

-  "Average winning margin was 7.7% in 2026 vs 11.8% in 2021 (valid votes, excl. NOTA)"
-  "Voters rejected the ruling party"

8. Chart playbook → TN 2026 research questions

RPC question	Primary chart	Secondary	Inspiration
Q1 Geographic	Regional stacked bars OR tile mosaic by region	Small multiples map	BBC, NPR tiles
Q2 Flips	Sankey (2021 party → 2026 party)	Highlight table of flipped ACs	AP VoteCast
Q3 TVK vote share	Marimekko or grouped bars by region	Dot plot statewide shares	538 composition charts
Q4 Reserved seats	Faceted bars GEN vs SC vs ST	Same metrics, side-by-side years	Hindu small-multiples
Q5 Turnout	Dumbbell top-20 ACs	Choropleth if turnout scraped	ECI portal + Quint live maps
Q6 Margins	Beeswarm/box/violin	KPI: % seats winner <35% share	FiveThirtyEight uncertainty framing

Recommended "creative hero" trio (data-backed):

1. **234-cell tile mosaic** (2026 winner, filter by region/flip)
2. **Sankey** seat flows 2021→2026
3. **Margin beeswarm** 2021 vs 2026 overlay

9. Tech stack decision matrix

Tool	Strengths	Weaknesses	Best for
Streamlit	Fast Python, filters, Plotly embed	Custom design limits	Hackathon dashboard + demo
Plotly / Dash	Sankey, interactivity, export	Dash learning curve	Sankey + drill-down
Tableau Public	Polish, sharing gallery	Repro on GitHub harder	Impressive static/interactive publish
Power BI	Corporate deck alignment	Less "creative web" feel	If reviewer expects Microsoft stack
D3.js / Observable	Maximum creativity	Time-intensive	Portfolio-grade custom viz
Flourish	Animated bars, Sankey templates	Hosted dependency	Quick hero GIF for LinkedIn
Jupyter + Matplotlib	Reproducibility	Less interactive	Notebook grading
GeoPandas + Folium/Plotly	Real maps	Boundary QA burden	Only with Datameet + caveats

Recommended stack for "creative + submittable":

```

pandas → processed CSVs
plotly → Sankey, mosaic, beeswarm
streamlit → multipage app (st.navigation)
kaleido → PNG exports for deck
jupyter → reproducibility

```

10. Dashboard information architecture (BBC + NPR + News18 hybrid)

10.1 Page structure (Streamlit multipage)

Page	User question	Key widgets
Overview	"What happened statewide?"	KPI row, headline, editorial rec card
Flow	"How did seats move?"	Sankey, flip count
Regions	"Where?"	Tile mosaic + regional bars
Fragmentation	"How close were races?"	Margin beeswarm, <35% table
Explorer	"Tell me about one AC"	Search <code>ac_number</code> , candidate table
Methods	"Can I trust this?"	Definitions, sources, limitations

10.2 UX best practices

- **Streamlit:** `st.set_page_config(layout="wide")`, `@st.cache_data`, filters in sidebar ([Streamlit multipage docs](#))
- **NPR:** Equal visual weight per constituency in grids
- **BBC:** Year toggle 2021 | 2026 everywhere
- **News18:** Search + sort on constituency table
- **Do not:** Auto-play sound; flashing colors; 3D pie charts

10.3 Visual design system (neutral)

```

Background:  #0f1419 (dark) or #f8fafc (light) – pick one theme, stay consistent
Text:        #e2e8f0 / #1e293b
Accent:      #3b82f6 (navigation only – not party-coded)
Party series: Okabe-Ito – assign arbitrarily by alphabet, document in README
Font:        Inter, Source Sans 3, or system-ui
Chart height: 480–600px for hero; 320px for small multiples

```

11. Data pipeline (all sources)

```

Starter pack (ac_number join)
  tn_2021_results.csv
  tn_2026_results.csv
  constituency_master.csv

Optional enrichment
  ECI results.eci.gov.in → 2026 turnout (blank in starter CSV)
  eci.gov.in/statistical-reports → audit PDFs
  datameet/maps → AC boundaries (with limitation footnote)
  indiavotes.com → spot checks

Derived tables (you create)
  ac_summary_2021.csv
  ac_summary_2026.csv
  ac_comparison.csv
  party_normalize.csv
  sankey_edges.csv

```

Metrics: See `docs/METRIC_DEFINITIONS.md` (create when building) — valid votes excl. NOTA, margin%, flip flag.

12. Creative build blueprint (end-to-end)

Phase A — Data (Day 1–2)

1. Copy raw → `data/raw/`
2. Build `party_normalize.csv` (top 30 raw labels)
3. Export `ac_comparison.csv` (234 rows)
4. Validate 2 ACs manually against ECI

Phase B — Hero visuals (Day 3–5)

1. **C1** Tile mosaic — Plotly scatter grid or custom CSS grid in Streamlit
2. **C2** Sankey — `go.Sankey` (Plotly docs)
3. **C3** Margin beeswarm — `px.strip` or violin by year
4. Export 1920×1080 PNG → `outputs/charts/`

Phase C — Dashboard (Day 5–7)

1. `streamlit run dashboard/app.py`
2. Multipage navigation + sidebar filters (region, reserved, flip)
3. `st.query_params` for shareable AC links (optional)

Phase D — Story package (Day 7–10)

1. Deck: hero = C1, slides 2–4 = C2–C3 + one regional bar
2. Video: screen-record dashboard + face cam
3. README: reproduction + neutrality disclaimer

Phase E — Submit (Day 10–12)

1. GitHub public
2. YouTube unlisted
3. LinkedIn (tags per `How_to_Submit.docx`)
4. RPC portal URL

13. Master submission checklist (everything)

Content quality

- ☐ One sentence headline works standalone
- ☐ 3 research questions answered with depth
- ☐ Editorial recommendation for 60-min show
- ☐ Limitations slide (Form-20, normalization, turnout)

Visual quality

- ☐ Hero visual understandable in 5 seconds
- ☐ 3 distinct chart types (not 3 bar charts)
- ☐ Accessibility checklist (Section 6)
- ☐ Neutrality tone audit (Section 7)

Technical

- ☐ Notebook runs clean env
- ☐ Dashboard runs via documented command
- ☐ Numbers match between deck, video, dashboard

Distribution

- ☐ Public GitHub
 - ☐ Video link
 - ☐ LinkedIn: `@codebasics`, Hemanand Vadivel, Dhaval Patel, Govt of Tamil Nadu
 - ☐ Hashtags: `#ResumeProjectChallenge #Codebasics #DataAnalytics #TamilNaduElection2026`
 - ☐ RPC page submission
-

14. Curated bibliography (bookmark list)

Maps & geography

- NPR tile maps: <https://blog.apps.npr.org/2015/05/11/hex-tile-maps>
- Vox on misleading maps: <https://www.vox.com/2016/6/2/11828628/election-maps-hard>
- Datameet AC maps: <https://github.com/datameet/maps/tree/master/assembly-constituencies>

Interactivity & charts

- Plotly Sankey: <https://plotly.com/python/sankey-diagram/>
- Plotly Sankey deep dive: <https://plotly.com/blog/sankey-diagrams>
- Python graph gallery Sankey: <https://python-graph-gallery.com/sankey-diagram-with-python-and-plotly/>

Journalism & storytelling

- Pudding structure: <https://www.storybench.org/pudding-structures-stories-visual-essays/>
- AP election role: <https://www.ap.org/elections/our-role/how-we-call-races>

Accessibility

- RSS styling guide: <https://royal-statistical-society.github.io/datavisguide/docs/styling.html>
- EU accessible palettes: <https://data.europa.eu/apps/data-visualisation-guide/accessible-colour-palettes>

India data

- ECI results: <https://results.eci.gov.in/>
- IIT Election Explorer: <https://info-design-lab.github.io/General-Election-Explorer/>
- Datameet india-election-data: <https://github.com/datameet/india-election-data>

Tools

- Streamlit multipage: <https://docs.streamlit.io/develop/concepts/multipage-apps/page-and-navigation>
- Information is Beautiful Awards: <https://www.informationisbeautifulawards.com/>

This project

- `HACKATHON_EXECUTION_PLAN.md`
- `input_files_for_participants_rpc/`

15. What “most creative” means for this RPC

Creativity $\neq$ more chart types. It means:

1. **One unforgettable hero** (tile mosaic or Sankey) that reviewers remember after the video

2. **Clear narrative arc** in the dashboard (Overview → Why it matters → Proof → Explorer)
3. **Dual encoding** (mosaic + table, map + bar) so accessibility and trust increase
4. **Honest limitations** surfaced in UI, not hidden in README footnotes
5. **Production polish** — consistent theme, no clutter, fast filters

Avoid: 3D pies, party-flag colors, leader images, causal headlines, prediction language.

Document version: 1.0 — consolidates global, India, creative, ethical, and RPC-specific references.